# Miracle Apata — Desktop Software Engineer

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## Summary

Experienced desktop software engineer specializing in cross-platform applications for **Windows** and **macOS** using **Python (PySide6)** and **SwiftUI**. Adept at building multilingual educational tools used by universities, academies, and institutions for standardized testing. Expert in packaging with PyInstaller and Inno Setup and delivering native desktop experiences.

## Work Experience

**DINATEN** — Santa Cruz de Tenerife, Spain
https://dinaten.com
**Aug 2023 – Present**

- Developed and maintained piXeval and **pixEBAP**, two educational desktop applications for managing and correcting multiple-choice exams.
- Built apps using **PySide6** (Qt for Python), compiled with **PyInstaller**, and packaged installers with **Inno Setup**.
- Created and maintained a **macOS-native version of piXeval** using **SwiftUI**, designed specifically for Apple users while ensuring feature parity with the Windows build.
- Ensured full cross-platform support across **Windows and macOS** with consistent UX.
- Supported multiple languages in the app interface: **Catalan**, **Spanish**, **English**, **Euskera**, and **French**.
- Integrated key features:
- Bulk import from Word, Excel, and Moodle
- Generation of up to 9 randomized models of the same exam
- Scanning and automatic grading of answer sheets (up to 30 exams/minute) using OCR and OMR techniques implemented with OpenCV2, TensorFlow, and other supporting libraries
- Delivery of student results via email
- Advanced statistics and competency evaluation

## Skills

- **Languages:** Python, Swift (SwiftUI), SQL
- **Frameworks & Libraries:** PySide6, SwiftUI, OpenCV2, TensorFlow

- **Tools & Packaging:** PyInstaller, Inno Setup, Git, macOS development tools
- **Platforms:** Windows, macOS
- **App Languages:** Spanish, English, Catalan, Euskera, French
- **Tech Focus:** OCR/OMR, automated exam grading, cross-platform application development

## Projects

- **piXeval**
- Download
- Intelligent exam correction and evaluation tool for teachers and institutions.
- **pixEBAP**
- Internal desktop application used for managing exam forms and student data.

## Personal Projects

- **SnapGrade** *(solo project, in active development)*
- Desktop app for generating, scanning, and grading multiple-choice exam sheets via OMR — no Scantron hardware, no proprietary format lock-in.
- Built the sheet-generation pipeline: auto-computed bubble-grid layouts (never too small to fill, never overflowing the page), barcode/QR-encoded sheet identity, and print-ready PDF export for single sheets or full batches.
- Built and calibrated the OMR grading engine with **OpenCV**: barcode/QR decoding resolves each sheet's identity and expected layout, bubbles are classified by measured ink-fill ratio rather than fixed guesses, and ambiguous answers are flagged for review instead of silently guessed.
- Implemented Excel-based roster import (prefilled sheets, ID-matched grading) and results export (summary + per-question detail) with **openpyxl**.
- Built the **PySide6** UI with a token-based theming system — five switchable themes generated from one shared QSS template.
- Sheet generation, the grading engine, and Excel I/O are built and validated against a calibration fixture; wiring the UI to those engines is the one remaining piece.

**Tech:** PySide6, OpenCV, pyzbar, qrcode, python-barcode, ReportLab, openpyxl, uv

## Education

**B.Eng. in Electrical & Electronics Engineering**
Federal University of Technology, Akure — Graduated 2024

## Languages

- English (Fluent, C1)
- Spanish (Conversational, B1)

## Contact


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